Mathematical Modelling For A Medicine: Aspirin

Payal Agarwal

India

SUMMARY

Aspirin is used in the treatment of mild to moderate pain, inflammation, and fever. It is also used as an antiplatelet agent to prevent myocardial infarction, stroke and transient ischemic episodes. Although aspirin is a very common medicine, it should not be used by everyone. The most common side effect from aspirin is stomach upset or indigestion. Taking aspirin with food lessens the chance of this side effect. The effect of aspirin begins 30 to 60 minutes after you take it. (Coated aspirin may need 1 to 8 hours to work.) The pain-relieving action of one dose usually lasts about 4 hours but may last up to 12 hours.

After the modelling cycle, we discussed the **quantity of aspirin in blood**. We started with our problem by constructing the mathematical model which is a **differential equation** to find the quantity of aspirin in blood at time t. We discussed **phase line** and the **stability** of its equilibrium solution. We implemented **Euler's Method** to a simplified model simulated the solutions. We found the right stepsize and estimated the error. We then looked at Euler's method for systems of $\mathbf{M_g}$ and $\mathbf{M_b}$. We determined equilibrium points of this system and constructed a **phase plane**.

We finally conclude the report by working on a specific problem: Multiple Drug Administration by considering different circumstances including the missed dosages and over dosage, determining a general formula that can be useful in creating Drug regimens for different people.

CONTENTS

List Of Variables

$M_g(t)$	Quantity of medicine in the GI tract in 'mg' at time ' **t** '
$M_b(t)$	Quantity of medicine in the blood in 'mg' at time ' **t** '
V_g	Volume of the contents in the GI tract in 'L'
V_b	Volume of blood of the person taking the medicine in 'L'
d_1	rate of diffusion with which medicine diffuses from GI tract to blood in $^{-1}$minute
d_2	rate of diffusion with which medicine diffuses from GI tract to blood $^{-1}$minute
M_{beq}	Quantity of medicine in the blood at equilibrium in 'mg'
C_g	Concentration of medicine in GI tract in 'mg/L'
C_p	Concentration of medicine in plasma (blood) in 'mg/L'
t	Time in minute and hour

Chapter 1: Introduction

French chemist, Charles Frederic Gerhardt, was the first to prepare acetylsalicylic acid (named aspirin in 1899) in 1853. **Aspirin**, or **acetylsalicylic acid (A.S.A.)**, is a salicylate drug, often used as an analgesics to relieve minor aches and pains, as an antipyretic to reduce fever, and as an anti-inflammatory medication. It is sometimes used to treat or prevent heart attacks, strokes, and chest pain (angina).Aspirin should be used for cardiovascular conditions only under the supervision of a doctor. This is called anti-clotting effect. Aspirin's popularity declined after the market releases of paracetamol (acetaminophen) in 1956 and ibuprofen in 1969.

Aspirin can increase the risk of bleeding in the stomach, small intestine, and brain. Normally, there is a layer that protects the insides of the stomach and intestine from the acid in your stomach. If aspirin is taken at high doses and for a long time, it can slowly damage this layer. This damage can lead to bleeding. Using aspirin to prevent blood clots can also affect the natural healing of damaged blood vessels and increase the risk of bleeding in the brain. Aspirin has been theorized to reduce cataract formation in diabetic patients, but one study showed it was ineffective for this purpose. An attempt of understanding the effectiveness of Aspirin at different dosages is made here so as to contribute to the conclusive study of its effects and extent of these effects on the person ingesting it.

CHAPTER 2: QUANTITATIVE STUDY

The introduction chapter showed that Aspirin is a promising drug but the consequential side effects deliver an unprotective reputation of the drug. Various factors are responsible for the interaction of Aspirin and hence it's effects. Aspirin is an acetyl derivative of salicylic acid that is a white, crystalline, weakly acidic substance, with melting point 135°C. Acetylsalicylic acid decomposes rapidly in solutions of ammonium acetate or of the acetates, carbonates, citrates or hydroxides of the alkali metals. Aspirin is known to interact with other drugs. Acetazolamide and ammonium chloride have been known to enhance the intoxicating effect of salicyclates, and alcohol also enhances the gastrointestinal bleeding associated with these types of drugs as well. Aspirin is known to displace a number of drugs from protein binding sites in the blood, including the anti-diabetic drugs tolbutamide and chlorpropamide, the immunosuppressant methotrexate, phenytoin, probenecid, valproic acid (as well as interfering with beta oxidation, an important part of valproate metabolism) and any nonsteroidal anti-inflammatory drug.

First, section 2.1 explores the recommended usage of various doses of Aspirin preferably distinguishing between adults and children. Section 2.2 discusses primarily about human toxicity, which occurs upon exceeding the ingestion of extra quantity of drug. Finally, based on the knowledge provided on dosage recommendations and extension, section 2.3 answers few of the commonly thought question with respect to overconsumption of aspirin.

Section 2.1: Dosage and Medication

- Fever:
    - Adult - 325 to 650 mg orally or rectally every 4 hours as needed, not to exceed 4 g/day
    - Pediatric -2 to 11 years: 10 to 15 mg/kg orally or rectally every 4 to 6 hours as needed, not to exceed 4 g/day.

      12 years or older: 325 to 650 mg orally or rectally every 4 hours as needed, not to exceed 4 g/day.

- Pain:
    - Adult - 325 to 650 mg orally or rectally every 4 hours as needed, not to exceed 4 g/day.
    - Pediatric- 2 to 11 years: 10 to 15 mg/kg orally or rectally every 4 to 6 hours as needed, not to exceed 4 g/day.

      12 years or older: 325 to 650 mg orally or rectally every 4 hours as needed, not to exceed 4 g/day.

- Rheumatic Fever:
    - Adult - 80 mg/kg/day orally in 4 equally divided doses, up to 6.5 g/day.
    - Pediatric - 90 to 130 mg/kg/day in equally divided doses every 4 to 6 hours, up to 6.5 mg/day.
    - Dosage may be adjusted according to patient response, tolerance, and serum salicylate levels (therapeutic range is 250 to 400 mcg/mL for rheumatic fever). Generally after 1 to 2 weeks, the dosage is decreased to approximately 60 to 70 mg/kg/day and given for an additional 1 to 6 weeks or longer if necessary, then gradually withdrawn over 1 to 2 weeks. An appropriate course of antibiotic therapy should be initiated at the time of diagnosis of rheumatic fever.

- Ischemic Stroke:
    - Adul - 50 to 325 mg orally once a day. Therapy should be continued indefinitely

❖ Myocardial Infarction:

- Adult - 160 to 162.5 mg orally once a day beginning as soon as an acute myocardial infarction is suspected and continuing for 30 days. If a solid dose formulation is used, the first dose should be chewed, crushed, or sucked. Long-term aspirin therapy for secondary prevention is recommended after 30 days.

❖ Angina Pectoris:

- Adult - 75 mg to 325 mg orally once a day beginning as soon as unstable angina is diagnosed and continuing indefinitely.

❖ Juvenile Rheumatoid Arthritis

- Pediatric - **2 to 11 years or less than or equal to 25 kg**:
  Initial: 60 to 90 mg/kg/day orally in equally divided doses.
  Maintenance: 80 to 100 mg/kg/day orally in equally divided doses; higher dosages, up to 130 mg/kg/day, may be necessary in some cases, not to exceed 5.4 g/day.

  12 years or older or greater than 25 kg:
  Initial: 2.4 to 3.6 g/day orally in equally divided doses.
  Maintenance: 3.6 to 5.4 g/day orally in equally divided doses; higher dosages may be necessary in some cases.

Liver Dose Adjustments

The use of aspirin in patients with severe hepatic impairment is not recommended due to the potential for increased risk of clinically significant bleeding and other adverse effects.

Section 2.2: Human Toxicity

Aspirin overdose can be acute or chronic. In acute poisoning, a single large dose is taken; in chronic poisoning, higher than normal doses are taken over a period of time. Acute overdose has a mortality rate of 2%. Chronic overdose is more commonly lethal, with a mortality rate of 25%.

Therapeutic dose of ASA in an adult is between 0.5 and 1.5 g/day. Accidental ingestions of <150 mg/kg can usually be treated at home with fluids and telephone follow-up. Accidental ingestion of 150 to 300 mg/kg (moderately severe overdose): if less than 1 hour after ingestion, induction of emesis and follow-up for 24 hours is suggested. Accidental ingestion of 300 – 500 mg/kg is considered as a severe overdose and requires hospitalization. Doses higher than 500 mg/kg are potentially lethal. The minimum lethal dose (MLD) was estimated to be 15 g/70 kg person.

Patients with mild to **moderate intoxication** may develop fever, tachypnea, tinnitus, respiratory alkalosis, metabolic acidosis, lethargy, mild dehydration, nausea and vomiting. At **severe intoxication**, encephalopathy, coma, hypotension, pulmonary edema, seizures, metabolic acidosis, coagulopathy (clot formation), cerebral edema and dysrhythmias, as well as kidney failure may be observed.

Therapeutic blood concentration for adults is between 20 and 100 mg/l, whereas toxic concentration is between 150 and 300 mg/l. The minimum lethal blood concentration is 500 mg/l. The mean lethal concentration, based on the data from several handbooks, was calculated to be 950 mg/l.

Section 2.3: Drug Adjustments

What happens if you miss a dose?

Since aspirin is used when needed, you may not be on a dosing schedule. If you are on a schedule, use the missed dose as soon as you remember. Skip the missed dose if it is almost time for your next scheduled dose. Do not use extra medicine to make up the missed dose.

What happens if you overdose?

Seek emergency medical attention. Initial treatment of an acute overdose includes gastric decontamination. This is achieved by administering activated charcoal, which adsorbs the aspirin in the gastrointestinal tract. Overdose symptoms may include temporary hearing loss, seizure (convulsions), or coma.

What should you avoid while taking aspirin?

Avoid drinking alcohol while you are taking aspirin. Alcohol may increase your risk of stomach bleeding.

If you are taking this medicine to prevent heart attack or stroke, avoid also taking **ibuprofen** (Advil, Motrin). Ibuprofen may make this medicine less effective. If you must use both medications, take the ibuprofen at least 8 hours before or 30 minutes after you take the aspirin (non-enteric coated form).

Ask a doctor or pharmacist before using any cold, allergy, or pain medication. Many medicines available over the counter contain aspirin or an NSAID. Taking certain products together can cause you to get too much of this type of medication. Check the label to see if a medicine contains aspirin, ibuprofen, ketoprofen, naproxen, or an NSAID.

CHAPTER 3: PHARMACOLOGY

In the previous chapter we were knowledged about the proper dose and their usage. We are now also aware about the overdose range for humans. Clearly, overdose of aspirin accounts for certain adverse effects on the health of the consumer. Initially, in the section 3.1 we determine the mechanism by which aspirin works. Followed by that, section 3.2 dictates the effects of aspirin on biological systems (here human metabolic activities and systems) - **Pharmacodynamics**. After that, section 3.3 talks about the effect of biological system on the drug – **Pharmacokinetics**.

It is followed by determination of different variables needed for the Basic Model.

Section 3.1: Mechanism of Action

Aspirin is **absorbed** rapidly from the stomach and intestine by passive diffusion. Aspirin is a prodrug, which is transformed into salicylate in the stomach, in the intestinal mucosa, in the blood and mainly in the liver. Salicylate is the active metabolite responsible for most anti-inflammatory and analgesic effects (but acetylsalicylate is the active moiety for the antiplatelet-aggregating effect). Gastrointestinal intolerance to salicylate observed in some patients has prompted the development of formulations with enteric coating. Salicylate distributes rapidly into the body fluid compartments. It binds to albumin in the plasma. With increasing total **plasma salicylate concentrations**, the unbound fraction increases. Salicylate may cross the placental barrier and distributes into breast milk.

As mentioned above, aspirin is rapidly **biotransformed** into the active metabolite, salicylate. Therefore, aspirin has a very short half-life. Salicylate, in turn, is mainly metabolized by the liver. This metabolism occurs primarily by hepatic conjugation with glycin or glucuronic acid, each involving different metabolic pathways. The predominant pathway is the conjugation with glycin, which is saturable. With low doses of aspirin approximately 90% of salicylate is metabolized through this pathway. As the maximum capacity of this major pathway is reached, the other pathways with a lower **clearance** become more important. Therefore, the **half-life** of salicylate depends on the major metabolic pathway used at a given concentration and becomes longer with increasing dosage. Salicylate is said to follow nonlinear kinetics at the upper limit of the dosing range. Studies have shown that there is much inter-subject variation with respect to the relative contribution of the different salicylate metabolic pathways.

Urinary excretion of unchanged salicylate accounts for 10% of the total elimination of salicylate. Excretion of salicylate results of glomerular filtration, active proximal tubular secretion through the organic acid transporters and passive tubular **reabsorption**. Urinary excretion is markedly pH dependant and as the urinary pH rises from 5 to 8, the amount of free ionized salicylate excreted increases from 3% of the total salicylate dose to more than 80% (by ion trapping in the urine). Salicylate metabolites are also excreted in the urine.

Section 3.2: Pharmacodynamics

Treatment of mild to moderate pain; fever; various inflammatory conditions; reduction of risk of death or MI in patients with previous infarction or unstable angina pectoris, or recurrent transient ischemia attacks (TIAs) or stroke in men who have had transient brain ischemia caused by platelet emboli.

Aspirin side effects:

Get emergency medical help if you have signs of an allergic reaction to aspirin: hives; difficult breathing; swelling of your face, lips, tongue, or throat.

Stop using this medicine and call your doctor at once if you have:

- ringing in your ears, confusion, hallucinations, rapid breathing, seizure (convulsions);
- severe nausea, vomiting, or stomach pain;
- bloody or tarry stools, coughing up blood or vomit that looks like coffee grounds;
- fever lasting longer than 3 days; or
- swelling, or pain lasting longer than 10 days.

Common aspirin side effects may include:

- upset stomach, heartburn;
- drowsiness; or
- mild headache.

Section 3.3: Pharmacokinetics

Absorption

- Rapidly and completely absorbed. T_{max} is 1 to 2 h (salicylic acid).

- Salicylate exhibits dose-dependent kinetics. Overdose may change pharmacokinetic parameters.

- *The plasma half-life* of ASA is 15-20 min, at the therapeutic dosage. The plasma half-life for salicylate is 2-3 h in low doses and up to 12 h at the higher doses. At overdose, the half-life may be up to 15-30 h.

- *Peak plasma concentration*: at the therapeutic doses, e.g. after 500 mg ingestion of ASA peak is achieved in 14 min; peak plasma concentration of salicylate is achieved in 0.5-1 h after ingestion. Time to peak for salicylate, after acute oral overdose, may be between 12 and 24 h.

- *Volume of distribution* (ASA): 0.2 l/kg.

- *Plasma protein binding*: 50-90% (ASA). Plasma salicylate is < 80% protein bound, especially to albumin.

- *The elimination half-life* of plasma salicylate is dose-depending and increases with higher doses. The reported half-lives in adults ranged between 2.4 and 19 h, at the doses of 0.25 g and 10-20 g of sodium salicylate, respectively.

- Salicylate may pass blood-brain barrier, but this process is very slow and restricted. However, it is easily passing placenta.

Distribution

- Widely distributed to all tissues and fluids, including CNS, breast milk, and fatal tissues.
-
- Approximately 90% of salicylate is protein bound at concentrations of less than 100 mcg/mL and approximately 75% is bound at concentrations of more than 400 mcg/mL.

Metabolism

Rapidly hydrolyzed to salicylic acid (active). Salicylic acid is conjugated in the liver to the metabolites.

Elimination

- Salicylic acid plasma half—life is approximately 6 h, but may exceed 20 h in higher doses. The half—life is approximately 15 to 20 min for aspirin.

- Elimination follows zero-order kinetics. Renal elimination of unchanged drug depends on urine pH. A pH of more than 6.5 increases renal Clearance of free salicylate from less than 5% to more than 80%.

CHAPTER 4: SIMPLIFIED MODEL

In this chapter the quantitative information provided in previous sections are used to determine the variables and a simplified model to start exploring deep into the interaction between the aspirin and body system of the human taking it. Firstly, in section 4.1 a Mathematical model for calculations is formulated. After that, section 4.2 constructs a phase line and determines equilibrium point for $\mathbf{M_b(t)}$. Then in section 4.3 $\mathbf{d_1}$ and $\mathbf{d_2}$ of the simplified equation is altered to check hoe sensitive results are to changes in parameters.

Section 4.1: Mathematical Model

Medication that you take orally, first dissolves in the contents of your stomach and intestines. Together those are called your ***gastrointestinal tract***, or ***GI*** tract for short. Then the medicine passes from the GI tract through the liver into the bloodstream. From the blood the medicine can reach the organ that needs it and do its work there. The kidney then clears the medicine out of the blood again.

- $\mathbf{M_g(t)}$: Quantity of medicine in the GI tract at time '$\mathbf{t}$', where the unit of time $\mathbf{t}$ is minutes and that of $\mathbf{Mg(t)}$ is mg
- $\mathbf{M_b(t)}$: Quantity of medicine in the blood at time '$\mathbf{t}$'
- $\mathbf{V_g}$: Volume of the contents in the GI tract, unit of $\mathbf{V_g}$ is Litres.
- $\mathbf{V_b}$: Volume of blood of the person taking the medicine in Litres.

In this model, the medicine passes from the GI tract to the blood (and from the blood to the kidney) by passive diffusion. So if the concentration of the medicine in the GI tract is higher than the concentration in the blood, the medicine diffuses with diffusion rate from the GI tract to the blood.

- $\mathbf{d_1}$: rate of diffusion with which medicine diffuses from GI tract to blood. Unit of $\mathbf{d_1}$ is amount of medicine that diffuses per unit time.
- $\mathbf{d_2}$: rate of diffusion with which medicine diffuses from GI tract to blood. Unit of $\mathbf{d_2}$ is amount of medicine that diffuses per unit time.

Average $\mathbf{V_g}$ of a healthy person of weight 120 to 160 pounds is found to be 6L, whereas $\mathbf{V_b}$ is known to be 5.5L.

$$\frac{dM_g(t)}{dt} = -d_1[\frac{M_g(t)}{V_g} - \frac{M_b(t)}{V_b}]$$

$$\frac{dM_b(t)}{dt} = d_1\left[\frac{M_g(t)}{V_g} - \frac{M_b(t)}{V_b}\right] - d_2\frac{M_b(t)}{V_b}$$

Simplified Medicine Model:

Taking the concentration of medicine in stomach $\mathbf{C_g} = \frac{\mathbf{M_g}}{\mathbf{V_g}}$ constant the model for $\mathbf{M_b(t)}$ reduces to:

$$\frac{dM_b(t)}{dt} = d_1\left[C_g - \frac{M_b(t)}{V_b}\right] - d_2\frac{M_b(t)}{V_b}$$

Section 4.2: Equilibrium Point, Phase Line

Note:

- We assume that $\mathbf{M_g(t)}$ is constant and equal to 325 mg. (A tablet of 325 mg dose is taken)
- $\mathbf{d_1}$ **is** amount of salt diffused per unit time from GI tract to blood

  Considering complete absorbtion of aspirin it takes 1 to 2 hours for diffusion of the active part of aspirin i.e. salicyclate by the blood, from the gastrointestinal tract, as a result of difference between drug concentrations of blood and GI tract. Therefore $\mathbf{d_1}$ can be determined by dividing the total amount of medicine diffused by total time taken by the blood to absorb it.

  Therefore $\mathbf{d_1}$ is approximated to be 5.41 of aspirin per minute.

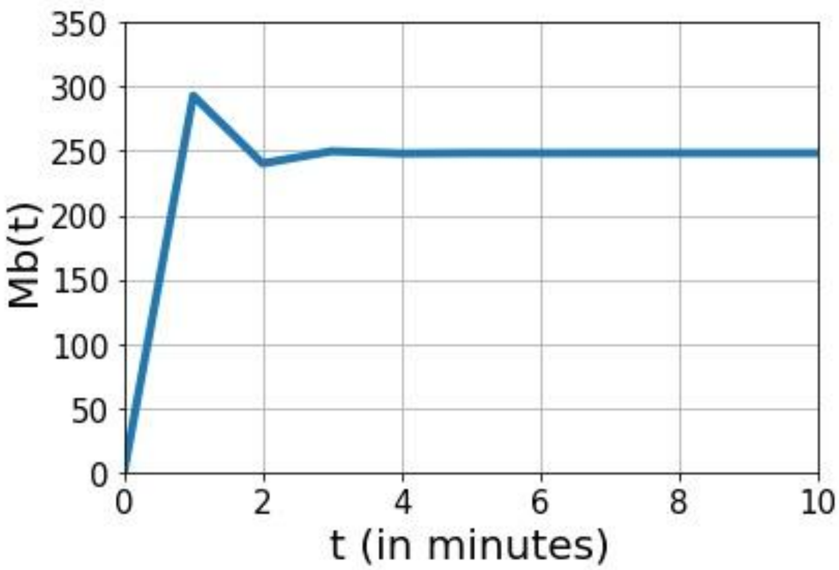
- $\mathbf{d_2}$ is amount of salt diffused per unit time from blood to kidney

  For a therapeutic dose it takes nearly 4 to 6 hours for the complete elimination of salicyclate (active part of aspirin) from blood to kidney due to the low plasma half-life of the drug. This becomes larger with higher dosage.

  Therefore $\mathbf{d_2}$ is approximated to be 1.08 aspirin per minute.

Implying the values mentioned above in the Simplified Mathematical model helps us to calculate the equilibrium point(s) for $\mathbf{M_b(t)}$**.** A graph(Fig 4.2.1) is also obtained after substituting respective values.

$$\frac{dM_b(t)}{dt} = d_1\left[C_g - \frac{M_b(t)}{V_b}\right] - d_2\frac{M_b(t)}{V_b}$$



4.2. 1: A graph of $M_b(t)$ vs time showing the slope $dM_b(t)/dt$

At equilibrium,

$\mathbf{M_b}$ is constant or same at any time t therefore rate of change in $\mathbf{M_b(t)}$ is equal to 0:

$$\frac{dM_b(t)}{dt} = 0$$

Therefore $\mathbf{M_{beq}}$ or quantity of blood at equilibrium can be calculated by the following equation:

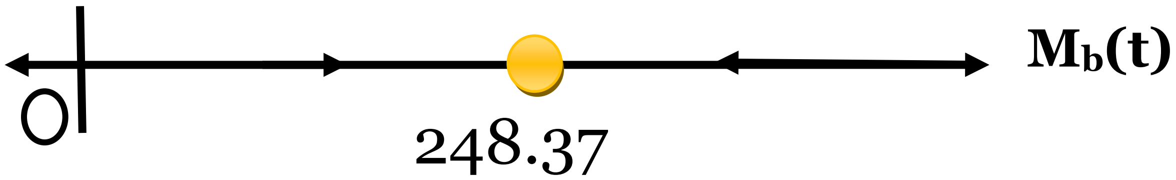
$$d_1\left[C_g - \frac{M_b(t)}{V_b}\right] - d_2\frac{M_b(t)}{V_b} = 0$$

$\mathbf{M_{beq} = 248.37}$ mg

Now in order construct a phase line, lets consider different initial conditions of $\mathbf{M_b(t)}$.

1. $M_b(0) = 0$, slope is $dM_b(0)/dt = 293.04$
2. $M_b(0) = 50$, slope is $dM_b(0)/dt = 234.04$
3. $M_b(0) = 100$, slope is $dM_b(0)/dt = 175.04$
4. $M_b(0) = 200$, slope is $dM_b(0)/dt = 57.04$
5. $M_b(0) = 300$, slope is $dM_b(0)/dt = -60.96$
6. $M_b(0) = 500$, slope is $dM_b(0)/dt = -296.96$

Therefore from the above data we can construct a phase line as follows:



4.2.2 : Phase line

This shows that if the $M_g(t)$ is constant then the amount of aspirin in the blood $M_b(t)$ will always try to reach the equilibrium concentration M_{beq} by either absorbing more drug from GI tract or eliminating it to kidney. Hence, the equilibrium so calculated can be called as **stable equilibrium**.

Section 4.3: Implement Euler's Method

For general differential equations

$$\frac{dy}{dt} = f(t, y)$$

the n-th step of Euler's Method is given by,

$$y\big((n+1)\Delta t\big) = y(n\Delta t) + \ \Delta t f(t, (y(n\Delta t))$$

Where Δt is the stepsize.

For our purpose, implementing Euler's Method would be,

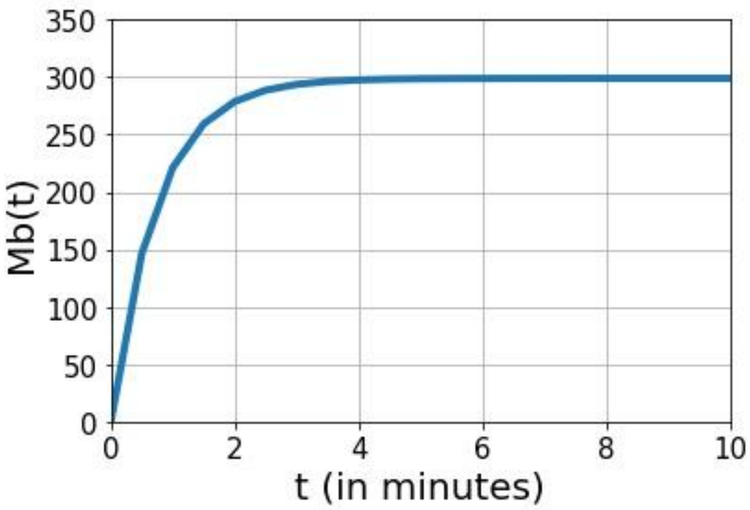
$$M_b\big((n+1)\Delta t\big) = M_b(n\Delta t) + \ \Delta t f(t, (M_b(n\Delta t))$$

To understand the sensitivity of parameters d_1 and d_2 w.r.t simplified model we can use Euler's method and accomplish calculation of error.

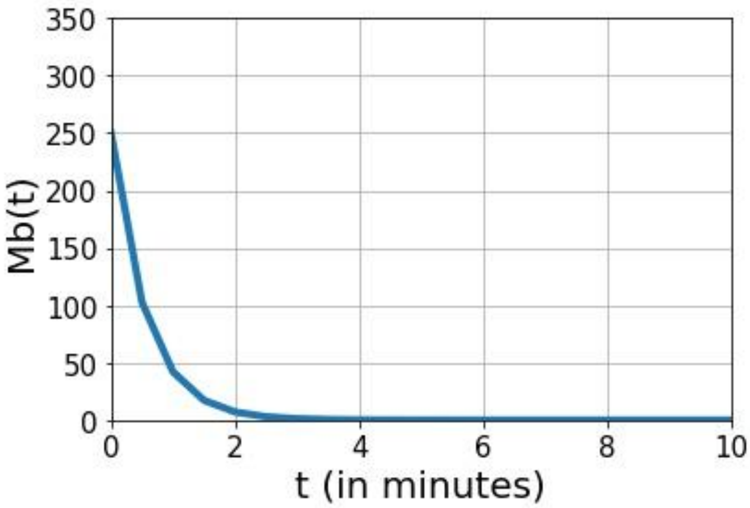
Case 1) **$d_2 = 0$; $d_1 = 5.41$; $M_b(0) = 0$**

Δt	$M_b(0.25)$	ERROR
1/4	73.26	
1/8	68.77	4.49
1/16	66.8	1.97
1/32	65.87	0.93
1/64	65.42	0.45

(An error of approx 0.50 mg in M_b is plausible. We calculate error around the steepest part of the slope $M_b(0.25)$.)



4.3.1 : Graph of $M_b(t)$ vs time t, when d_2 is taken to be 0.



4.3.2 : Graph of $M_b(t)$ vs time t, when d_1 is taken to be 0.

Case 2) **$d_1 = 0$; $d_1 = 1.18$; $M_b(0) = 250$**

Δt	$M_b(0.25)$	ERROR
1/4	176.25	
1/8	181.69	-5.44
1/16	184.01	-2.32
1/32	185.09	-1.08
1/64	185.59	-0.5

From the two graphs above 4.3.1 and 4.3.2 , it can be comprehended that when $\mathbf{d_2 = 0}$, after a certain time the quantity of aspirin present in blood remains constant even though the rate of diffusion of aspirin from blood to kidney is 0. Whereas, when $\mathbf{d_1 = 0}$ the graph decreases and finally remains 0 over time. This compels the elimination of aspirin from the blood completely after $\mathbf{M_b}$ at its peak plasma concentration or any other concentration.

Experiment with different combinations of $\mathbf{d_1}$ and $\mathbf{d_2}$ were performed.

CHAPTER 5: BASIC MODEL

We now consider both $\mathbf{M_g(t)}$ and $\mathbf{M_b(t)}$ as a system. On the previous chapter we have seen an initial value problem of $\mathbf{M_b(t)}$. We shall now learn about both the value as a system using the following equations.

$$\frac{dM_g(t)}{dt} = -d_1[\frac{M_g(t)}{V_g} - \frac{M_b(t)}{V_b}]$$

$$\frac{dM_b(t)}{dt} = d_1\left[\frac{M_g(t)}{V_g} - \frac{M_b(t)}{V_b}\right] - d_2\frac{M_b(t)}{V_b}$$

- $\mathbf{d_2 = 5.41}$ amount of aspirin per minute
- $\mathbf{d_1 = 1.08}$ amount of aspirin per minute
- $\mathbf{V_g = 6}$ Litre
- $\mathbf{V_b = 5.5}$ Litre
- $\mathbf{M_g(0) = 325}$ mg
- $\mathbf{M_b(0) = 0}$ mg

Section 5.1: Implement Euler's Method to system

For the general differential equations

$$\frac{d\vec{X}}{dt} = f(t, \vec{X})$$

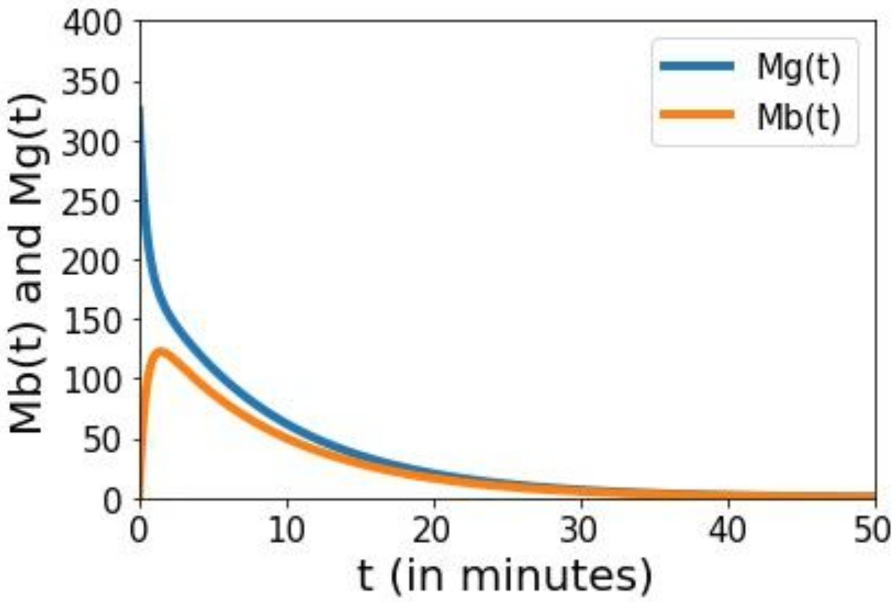
the n-th step of Euler's Method is given by,

$$\vec{X}\big((n+1)\Delta t\big) = \vec{X}(n\Delta t) + \Delta t f(t, (\vec{X}(n\Delta t))$$

Where $\mathbf{\Delta t}$ is the stepsize.

For systems of differential equations, it is convenient to combine the different dependent variables in a vector.

$$\vec{X}(t) = \begin{pmatrix} M_b(t) \\ M_g(t) \end{pmatrix}$$



5.1.1 . A graph representing Mb (t) and Mg(t) vs time t as a system

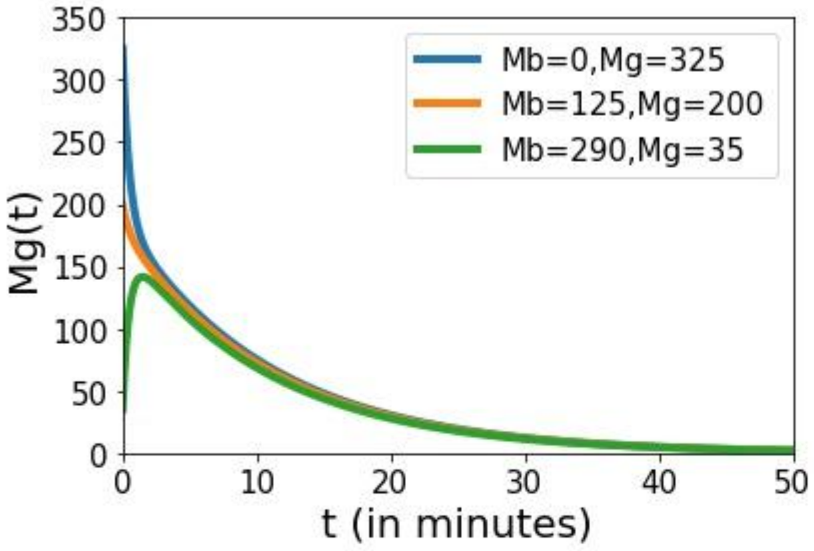
After double checking the errors in $\mathbf{M_b}$ the following table was observed

Δt	$M_b(0.25)$	ERROR
1/4	73.125	
1/8	63.39	4.49
1/16	59.71	1.97
1/32	58.02	0.93
1/64	57.54	0.49

This implyies that the error calculated are precisely plausible.

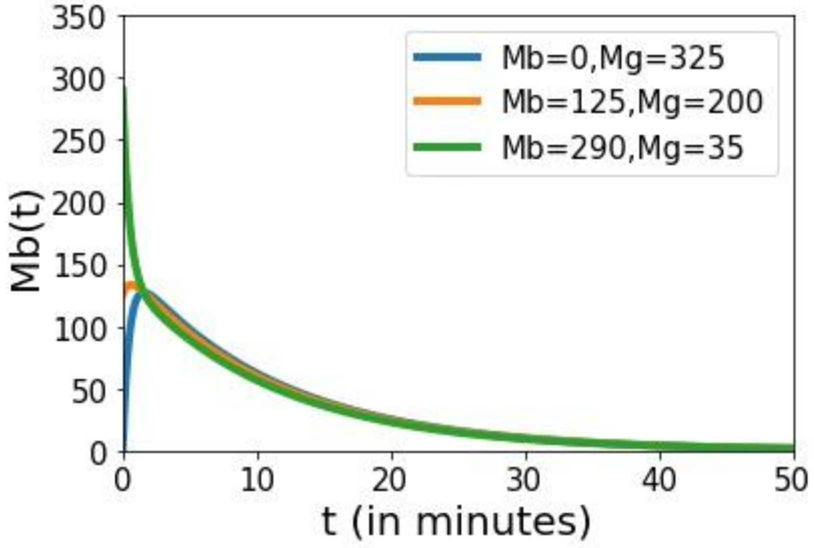
Section 5.2: Phase Plane

- First a graph of $M_g(t)$ vs t is plotted, where different initial values of $M_g(0)$ and $M_b(0)$ are considered as follows:



5.2. 1 Graph of Mg(t) vs time t

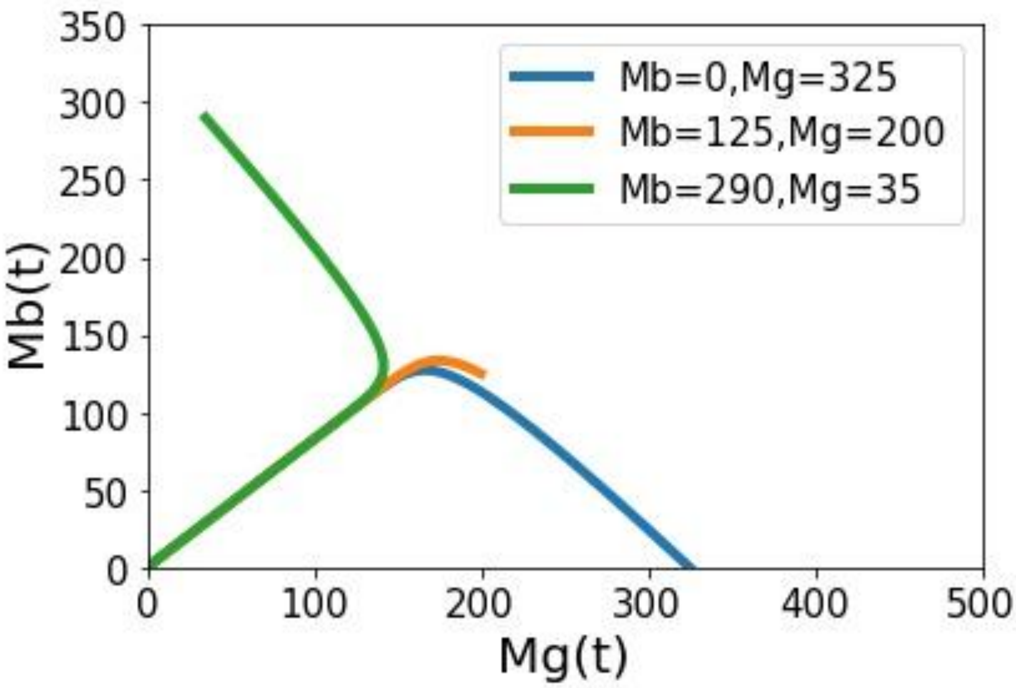
- Similarly, First a graph of $M_b(t)$ vs t is plotted:



5.2. 2 Graph of Mb(t) vs time t

No matter what initial conditions are taken, after consumption of 1 325mg tablet of aspirin the effects wear off soon as shown in the graph.

- Now we plot the different solutions for different sets of initial condition are plotted in **phase plane**, the $\mathbf{M_g M_b}$ **-** plane.



5.2. 3 Graph of Mb(t) vs time

CHAPTER 6: MULTIPLE DOSAGE REGIMEN

A single dose may provide an effective treatment. But the duration of illness is longer than the therapeutic effect produced by a single dose. In such cases drugs are required to be taken on a repetitive basis over a period of time.
The multiple dosing achieves and maintains drug concentration in plasma or at the site of action that are both safe and effective

Two major parameters that can be adjusted in developing a dosage regimen are:
1. The dose size: - It is the quantity of the drug administered each time.
2. Dose frequency: - It is the time interval between doses.

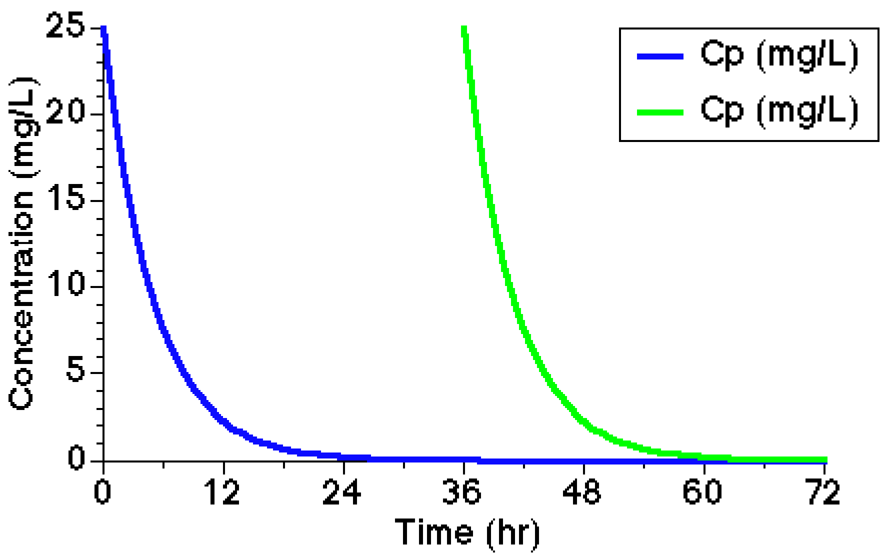
based on the pharmacokinetics and other equations we determine in the previous chapters , here in section 6.1 we will learn about drug accumulation problem. After that in section 6.2 we determine basic equation for multiple Aspirin dosage administration. In the next section of this chapter 6.3 we determine $\mathbf{C_{pmax}}$ and $\mathbf{C_{pmin}}$. Last section 6.4 specifically helps to determine the maintenance of Oral Dosage Administration.

Section 6.1: Drug Accumulation

After a single dose administration we assume that there is no drug in the body before the drug is administration and that no more is going to be administered. However, in the case of multiple dose administration we are expected to give second and subsequent doses before the drug is completely eliminated. Thus ACCUMULATION of the drug should be considered. On repeated drug administration the plasma concentration will be repeated for each dose interval giving a PLATEAU or STEADY STATE with the plasma concentration fluctuating between a minimum and maximum value.

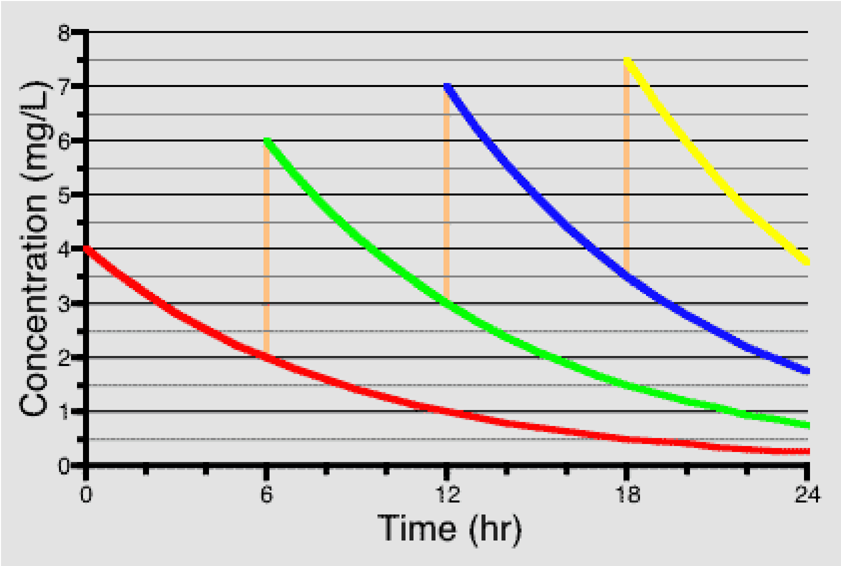
C_p : Concentration of medicine in plasma (blood) in 'mg'

If the doses are given far enough apart then the concentration will have fallen to approximately zero before the next dose. There will then be no accumulation of drug in the body.



6. 1.1 Drug concentration in plasma or blood after two independent dosese

However if the second dose is given early enough so that not all of the first dose is eliminated then the drug will start to accumulate and we will get higher concentrations with the second and third dose.



6.1 .2 Linear Plot of Cp versus Time Showing Doses Every Six Hours

With each dose, drug accumulated until the amount of drug eliminated during each dosing interval was equal to the amount of the dose. In the first interval plasma concentrations fall from 4 to 2 mg/L. Continuing for a number of doses gives the following table.

Table: Drug Concentration Accumulation During Multiple Aspirin Dose Administration

Start		End	Concentration lost during dosage interval
4	-->	2 mg/L	2 mg/L
6	-->	3	3
7	-->	3.5	3.5
7.5	-->	3.75	3.75
	...		
8	-->	4	4 <- which is the same as the concentration increase caused by each dose

There is a limit to drug accumulation because as the plasma concentration increased the amount of drug eliminated during the dosing interval will also increase as the rate of elimination is equal to the amount of the drug in the body multiplied by the rate constant for a first order elimination. (Compare this with the case of a continuous infusion).

So far we can see that if we give repeated doses before the body can eliminate the previous doses then we will get accumulation of the drug. We have also seen that when we have first order elimination this accumulation will not proceed indefinitely but will level off.

DRUG ACCUMULATION

A) Drug accumulation during multiple dosing: following the 1^{st} dose, if the 2^{nd} dose is given early enough so that not the entire 1^{st} dose is eliminated then the drug will start accumulating and we will get higher concentration with the 2^{nd} and 3^{rd} dose.

B) Steady state during multiple dosing:

The suitable amount of dose and identical dosing interval leads to steady state at which the mass of drug administered or absorbed is equal to the mass of drug eliminated

Section 6.2: Determination of General Equation

Describe the plasma concentration at any time after multiple Aspirin drug administration.

Concentration at the end of the first dosing interval

$$C_{p_1}^{\tau} = C_{p_1}^{0} * e^{-kel*\tau}$$

Cp after the First Dose

Where

$$Cp_{\text{dose number}}^{\text{time since last dose}}$$

This gives the plasma concentrations at the end of first interval, where τ is the dosing interval in hours.

Concentration at the start of the second interval

$$Cp_2^0 = Cp_1^{\tau} + Cp_1^0 = Cp_1^0 \bullet e^{-kel \bullet \tau} + Cp_1^0$$

Cp at the Start of the Second Interval

Concentration at the end of the second dose interval

$$Cp_2^{\tau} = \left[Cp_1^0 \bullet e^{-kel \bullet \tau} + Cp_1^0\right] \bullet e^{-kel \bullet \tau}$$

It will help if we define the parameter, $R = e^{-kel \cdot \tau}$, which is the fraction of the initial plasma concentration remaining at the end of the dosing interval. Then

$$Cp_2^{\tau} = Cp_1^0 \bullet R + Cp_1^0 \bullet R^2$$

$$Cp_3^0 = Cp_1^0 + Cp_1^0 \bullet R + Cp_1^0 \bullet R^2$$

Cp at the Start of the Third Interval

This is a geometric series with each term R times the preceding term.

$$Cp_n^0 = Cp_1^0 + Cp_1^0 \bullet R + Cp_1^0 \bullet R^2 + \ldots + Cp_1^0 \bullet R^{n-1}$$

Cp at the Start of the nth Interval

$$Cp_n^\tau = Cp_1^0 \bullet R + Cp_1^0 \bullet R^2 + Cp_1^0 \bullet R^3 + \ldots + Cp_1^0 \bullet R^n$$

These two sums are sums of geometric series they can be simplified to give ($V=V_b$)

$$Cp_n^0 = Cp_1^0 \bullet \left[\frac{1-R^n}{1-R}\right] = \frac{Dose}{V} \bullet \left[\frac{1-e^{-n\bullet kel\bullet\tau}}{1-e^{-kel\bullet\tau}}\right]$$

Cp at the Start of the nth Interval

$$Cp_n^\tau = Cp_1^0 \bullet \left[\frac{1-R^n}{1-R}\right] \bullet R = \frac{Dose}{V} \bullet \left[\frac{1-e^{-n\bullet kel\bullet\tau}}{1-e^{-kel\bullet\tau}}\right] \bullet e^{-kel\bullet\tau}$$

Cp at the End of the nth Interval

Using the first equation we can calculate drug concentration in blood or plasma at any time following uniform multiple Aspirin administration.

$$Cp_n^t = Cp_n^0 \bullet e^{-kel\bullet t}$$

Cp at time,t, after the nth Aspirin Dose

Where t = time since the last dose.

$$Cp_n^t = \frac{Dose}{V} \bullet \left[\frac{1-e^{-n\bullet kel\bullet\tau}}{1-e^{-kel\bullet\tau}}\right] \bullet e^{-kel\bullet t}$$

Cp at time,t, after the nth Aspirin Dose General Equation

Section 6.3: C_{pmax} and C_{pmin}

More useful equations can be derived from this general equation. These are equations to calculate the maximum and minimum plasma concentration after many doses. That is as n ---> ∞, R^n (= $e^{-n \bullet kel \bullet \tau}$) ---> 0 and with t = 0 or t = τ. These are the limits of the PLATEAU CONCENTRATIONS.

$$Cp_{\infty}^{0} = Cp_{max} = \frac{Dose}{V} \bullet \left[\frac{1}{1 - e^{-kel \bullet \tau}}\right] = \frac{Dose}{V \bullet (1 - R)}$$

$$Cp_{\infty}^{\tau} = Cp_{min} = \frac{Dose}{V} \bullet \left[\frac{e^{-kel \bullet \tau}}{1 - e^{-kel \bullet \tau}}\right] = \frac{Dose \bullet R}{V \bullet (1 - R)}$$

Cp Immediately before Many Doses

An example may be helpful: $t_{1/2}$ = 4 hr; IV dose 100 mg every 6 hours; V = 10L

$$C_{p_1}^{\ 0} = \frac{100}{10} = 10\ mg/L$$

What are the C_{pmax} and C_{pmin} values when the plateau values are reached

$$kel = \frac{0.693}{4} = 0.17/hr$$

$$R = e^{-kel * \tau} = e^{-0.17 * 6} = 0.35$$

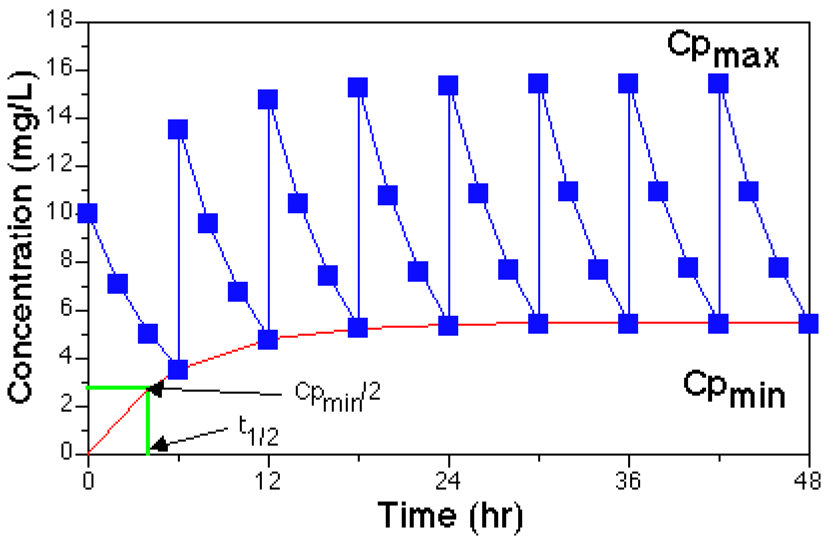
Therefore

$$Cp_{max} = \frac{Cp_1^0}{(1 - R)} = \frac{10}{1 - 0.35} = 15.5\ \mathrm{mg/L}$$

$$Cp_{min} = \frac{Cp_1^0 \bullet R}{(1 - R)} = Cp_{max} \bullet R = 15.5 \times 0.35 = 5.4\ \mathrm{mg/L}$$

Therefore the plasma concentration will fluctuate between 15.5 and 5.4 mg/liter during each dosing interval when the plateau is reached.

We can now calculate the plasma concentration at any time following multiple IV bolus administration and we can calculate the Cp_{max} and Cp_{min} values.



6.3.1 Plot of Cp Versus Time showing Time to Approach 50% of Plateau during Multiple Dose Regimen

It can be shown that the time to reach a certain fraction of the plateau concentration is dependent on the drug elimination half-life only, much the same as for the approach to steady state during an IV infusion. Thus we may have a problem with an excessive time required to reach the plateau. Therefore we may want to determine a suitable loading dose to achieve steady state rapidly.

In the previous example Cp_{max} = 15.5 mg/liter

A suitable loading dose would be $Cp_{max} * V$

155 mg as a bolus would give Cp = 15.5 mg/liter, followed by 100 mg every 6 hours to maintain the Cp_{max} and Cp_{min} values at 15.5 and 5.5 mg/liter respectively.

In summary:

The IV bolus loading dose to quickly achieve a required drug concentration, Cp_{max}, can be calculated as $Cp_{max} * V$

Rewriting Equation with Dose expressed more explicitly as the Maintenance Dose.

$$Cp_{max} = \frac{\text{Maintenance Dose}}{V \bullet (1 - R)}$$

$$\text{Loading Dose} = \frac{\text{Maintenance Dose}}{(1 - R)}$$

$$\text{Maintenance Dose} = \text{Loading Dose} \bullet (1 - R)$$

We can try another example of calculating a suitable dosing regimen.

Consider V = 25 liter; kel = 0.15 hr^{-1} for a particular drug and we need to keep the plasma concentration between 35 mg/liter (MTC) and 10 mg/liter (MEC).

What we need is the maintenance dose, the loading dose, and the dosing interval.

$$\frac{Cp_{max}}{Cp_{min}} = \frac{Dose}{V \bullet (1 - R)} \bullet \frac{V \bullet (1 - R)}{Dose \bullet R} = \frac{1}{R}$$

Therefore

$$R = \frac{Cp_{min}}{Cp_{max}} = \frac{10}{35} = 0.286 = e^{-kel \bullet \tau}$$

Taking the ln of both sides

$$-kel * \tau = -1.252 = -0.15 * \tau$$

τ (dosing interval) = 8.35 hr

A dosing interval of 8 hours would be more reasonable. Thus with τ = 8 hr and kel = 0.15 hr^{-1}

$R = e^{-kel \bullet \tau} = e^{-8 \times 0.15} = 0.301$

$$\text{Maintenance Dose} = Cp_{max} \bullet V \bullet (1 - R)$$

Maintenance dose = 35 x 25 x (1 - 0.301) = 612 mg

Again a more realistic maintenance dose would be 600 mg every 8 hours.

To check this regimen

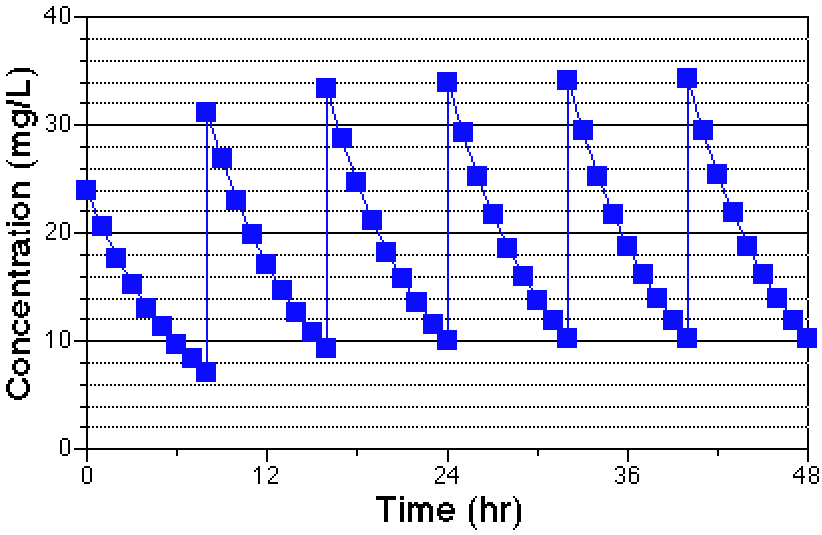
$$Cp_{max} = \frac{Dose}{V \bullet (1 - R)} = \frac{600}{25 \times (1 - 0.301)} = 34.3\ mg/L$$

$$Cp_{min} = Cp_{max} \bullet R = 34.3 \times 0.301 = 10.3\ mg/L$$

This regimen would be quite suitable as the maximum and minimum values are still within the limits suggested. All that remains is to calculate a suitable loading dose.

Loading dose = $Cp_{max} \bullet V$ = 35 x 25 = 875 mg either 875, 850 or 800 mg

This answer can be expressed graphically.



6.3.2 Plasma Concentration after Multiple IV Bolus Doses

Section 6.4: Multiple Oral Dose Administration

To start the plasma concentration achieved following a single oral dose can be given by:

$$Cp = \frac{F \bullet Dose \bullet ka}{V \bullet (ka - kel)} \bullet \left[e^{-kel \bullet t} - e^{-ka \bullet t}\right]$$

Cp after a Single Oral Dose

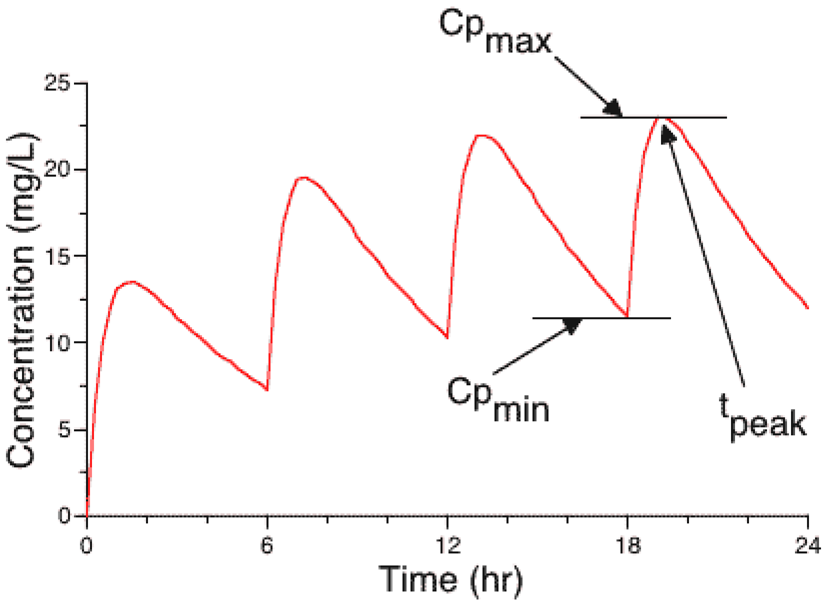
This can be converted to an equation describing plasma concentration at any time following n equal doses with constant dosing interval t using a "multiple dose function".

$$e^{-k \bullet t} \rightarrow \left[\frac{1 - e^{-n \bullet k \bullet \tau}}{1 - e^{-k \bullet \tau}}\right] \bullet e^{-k \bullet t}$$

Multiple Dose Function

$$Cp = \frac{F \bullet Dose \bullet ka}{V \bullet (ka - kel)} \bullet \left(\left[\frac{1 - e^{-n \bullet kel \bullet \tau}}{1 - e^{-kel \bullet \tau}}\right] \bullet e^{-kel \bullet t} - \left[\frac{1 - e^{-n \bullet ka \bullet \tau}}{1 - e^{-ka \bullet \tau}}\right] \bullet e^{-ka \bullet t}\right)$$

Cp after a Single Oral Dose - Uniform Dose and Interval, τ
<u>General Equation</u>



6.4.1 Plot of Cp Versus Time for Multiple Oral Doses showing Cp_{max} and Cp_{min}

The plasma concentration versus time curve described by this equation is similar to the Aspirin curve in that there is accumulation of the drug in the body to some plateau level and the plasma concentrations fluctuate between a minimum and a maximum value.

The Cpmax value could be calculated at the time $t = t_{peak}$ after many doses ($n \rightarrow \infty$) but it is complicated by the need to determine the value for t_{peak}.

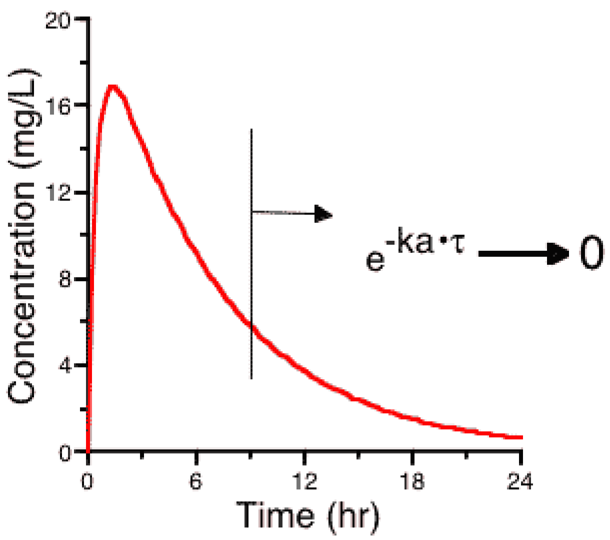
Cp_{min} Equation

However Cp_{min} can be more easily determined at t = 0 or t = t. Thus at t = 0 and $n \rightarrow \infty$ as $e^{-n \cdot k \cdot \tau} \rightarrow 0$.

$$Cp_{min} = \frac{F \bullet Dose \bullet ka}{V \bullet (ka - kel)} \bullet \left(\left[\frac{1}{1 - e^{-kel \bullet \tau}}\right] - \left[\frac{1}{1 - e^{-ka \bullet \tau}}\right]\right)$$

Cp_{min} after Many Oral Doses

This can be further simplified if we assume that the subsequent doses are given after the plasma concentration has peaked and $e^{-ka \cdot \tau}$ is close to zero. That is the next dose is given after the absorption phase is complete.



6.4.2 Plot Cp Versus Time after a Single Dose showing Possible Time of Second Dose

Cp_{min} then becomes:

$$Cp_{min} = \frac{F \bullet Dose \bullet ka}{V \bullet (ka - kel)} \bullet \left[\frac{e^{-kel \bullet \tau}}{1 - e^{-kel \bullet \tau}}\right]$$

Cp_{min} after Many Oral Doses - Version 2

The relationship between loading dose and maintenance dose and thus drug accumulation during multiple dose administration can be studied by looking at the ratio between the minimum concentration at steady state and the concentration one dosing interval,τ, after the first dose. [Assuming $e^{-ka \cdot \tau}$ is close to zero].

$$\frac{Cp_{min}}{Cp_1^{\tau}} = \frac{\frac{F \bullet Dose \bullet ka}{V \bullet (ka - kel)} \bullet \left[\frac{e^{-kel \bullet \tau}}{1 - e^{-kel \bullet \tau}}\right]}{\frac{F \bullet Dose \bullet ka}{V \bullet (ka - kel)} \bullet e^{-kel \bullet \tau}}$$

Ratio between Cp after First and Last Dose

Which can be simplified to give:-

$$\frac{Cp_{min}}{Cp_1^{\tau}} = \frac{1}{1 - e^{-kel \bullet \tau}} = \frac{1}{(1 - R)}$$

Ratio between Cp after First and Last Dose

This turns out to be the same equation as for the Aspirin. Therefore we can calculate a loading dose just as we did for an Aspirin multiple dose regimen.

$$\text{Loading Dose} = \frac{\text{Maintenance Dose}}{(1 - R)}$$

Equation 26.2.5 Loading Dose Equation

This equation holds if each dose is given after the absorption phase of the previous dose is complete.

We can further simplify Equation if we assume that ka >> kel then (ka - kel) is approximatley equal to ka and ka/(ka - kel) is approximately equal to one.

$$Cp_{min} = \frac{F \bullet Dose}{V} \bullet \left[\frac{e^{-kel \bullet \tau}}{1 - e^{-kel \bullet \tau}}\right]$$

Equation 26.2.6 Cp_{min} after Many Oral Doses - Version 3

This Equation is an even more extreme simplification. However, it can be very useful if we don't know the ka value but we can assume that absorption is reasonably fast. This equation will tend to give concentrations that are lower than those obtained with the full equation. Thus any estimated fluctuation between Cp_{min} and Cp_{max} will be overestimated using the simplified equation.

$\overline{Cp}$ Equation:

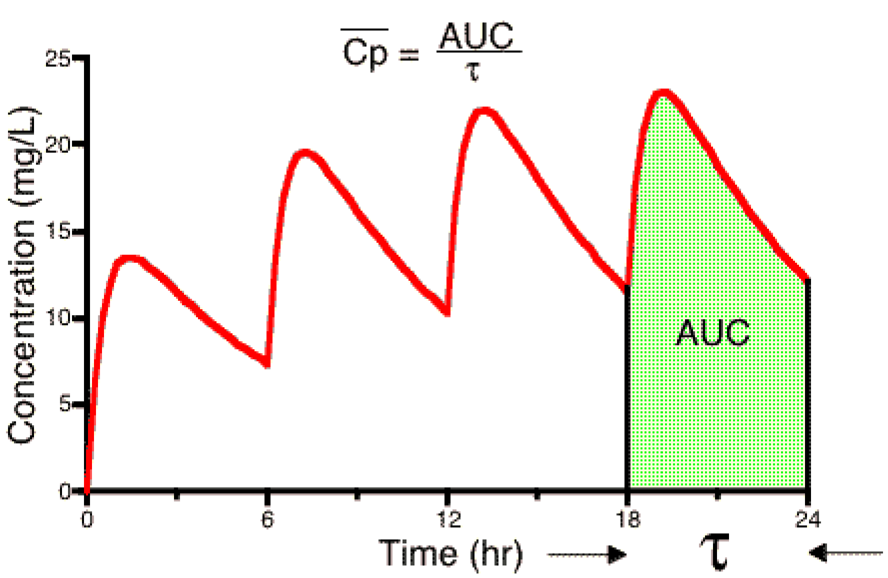
Another very useful concentration term for the calculation of oral dosing regimens is the average plasma concentration, $\overline{Cp}$, during the dosing interval at steady state.

This term is defined as the area under the plasma concentration versus time curve during the dosing interval at steady state divided by the dosing interval.

Thus:

$$\overline{Cp} = \frac{AUC}{\tau} = \frac{\int_0^{\tau} Cp \bullet dt}{\tau}$$

Average Cp for a Dosing Interval at Steady State



6.4.3 Plot of Cp versus Time after Multiple Oral Administration showing AUC at Steady State

Since,

$$AUC = \frac{F \bullet Dose}{kel \bullet V}$$

$$\overline{Cp} = \frac{F \bullet Dose}{V \bullet kel \bullet \tau}$$

Average Cp for a Dosing Interval at Steady State

Note that the AUC during one dosing interval at steady state is the same as the AUC from zero to infinity after one single dose.

Also, an interesting result of this equation is that we get the same average plasma concentration whether the dose is given as a single dose every dosing interval, τ, or is subdivided into shorter dosing intervals. For example 300 mg every 12 hours will give the same average plasma concentration as 100 mg every 4 hours. **However**, the difference between the maximum and minimum plasma concentration will be larger with less frequent dosing.

An Example - Part 1

With F = 1.0; V = 30 liter; $t_{1/2}$ = 6 hours or kel = 0.693/6 = 0.116 hr^{-1}, calculate the dose given every 12 hours that will achieve an average plasma concentration of 15 mg/L.

$$\text{Since } \overline{Cp} = \frac{F \bullet Dose}{V \bullet kel \bullet \tau}$$

$$Dose = \frac{\overline{Cp} \bullet V \bullet kel \bullet \tau}{F} = \frac{15 \times 30 \times 0.116 \times 12}{1.0} = 624\ mg$$

We could now calculate the loading dose

$$R = e^{-kel \bullet \tau} = e^{-0.166 \times 12} = 0.25$$

$$\text{Loading Dose} = \frac{\text{Maintenance Dose}}{1 - R} = \frac{624}{1 - 0.25} = 832\ mg$$

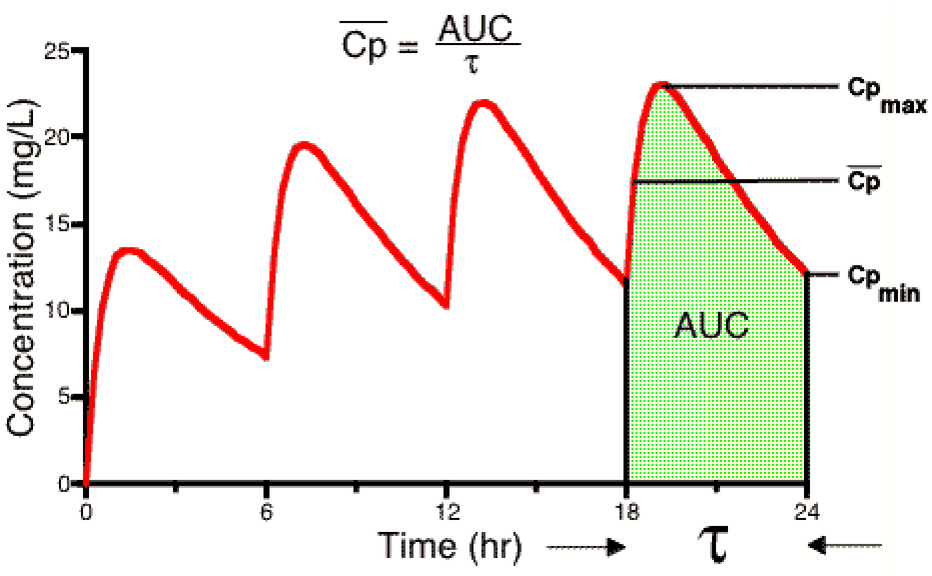
To get some idea of the fluctuations in plasma concentration we could calculate the Cpmin value.

Assuming that ka >> kel and that $e^{-ka \bullet \tau}$ --> 0, using Equation 26.2.6.

$$Cp_{min} = \frac{F \bullet Dose}{V} \bullet \left[\frac{e^{-kel \bullet \tau}}{1 - e^{-kel \bullet \tau}} \right]$$

$$Cp_{min} = \frac{1.0 \times 624}{30} \times \left[\frac{0.25}{1 - 0.25} \right] = 6.93\ mg/L$$

Therefore the plasma concentration would probably fluctuate between 7 and 23 mg/L (very approximate) with an average concentration of about 15 mg/L. [23 = 15 + (15 - 7), i.e. high = average + (average - low), very approximate!].



6.4.4 Figure Illustrating Cp_{max}, Cp_{min} and Cp(average)

An Example - Part 2

As an alternative we could give half the dose, 312 mg, every 6 hours to achieve:

$$Cp_{min} = \frac{1.0 \times 312}{30} \times \left[\frac{0.5}{1 - 0.5}\right] = 10.4\ mg/L$$

The $\overline{Cp}$ would be the same

$$\overline{Cp} = \frac{F \bullet Dose}{V \bullet kel \bullet \tau} = \frac{1 \times 312}{30 \times 0.116 \times 6} = 15\ mg/L$$

Thus the plasma concentration would fluctuate between about 10.4 to 20 with an average of 15 mg/L.

BIBLIOGRAPHY

1. Poisindex, Thomson Micromedex (2005).
2. Ellenhorn, M.J.& Barceloux, D.G. (1988) *Medical Toxicology. Diagnosis and Treatment of Human Poisoning,* pp. 156-168. New York: Elsevier.
3. Kaye, S. (1980) *Handbook of Emergency Toxicology: A Guide for the Identification, Diagnosis and Treatment of Poisoning*, 4th edn., pp. 152-156, Springfield, IL, USA.
4. Winek, C.L. (1994) Drug and chemical blood-level data. *Winek's Toxicological Annual,* Pittsburgh. Allegheny County Department Laboratories.
5. Ekwall, B., Clemedson, C., Crafoord, B., Ekwall, B., Hallander, S., Walum, E., Bondesson, I. (1998) MEIC evaluation of acute systemic toxicity. Part V. Rodent and human toxicity data for the 50 reference chemicals. ATLA 26, 571-616.
6. *Therapeutic Drugs* (1993) Vol 1 & 2, Dollery C, ed., London: Churchill Livingstone.
7. Goodman Gilman, A., Rall, T.W., Nies, A.S. & Tylor, P., eds. (1990) *The Pharmaceutical Basis of Therapeutics*. 8th edn., New York: Pergamon Press.
8. Baselt, R.C. & Cravey, R.H. (1995) Disposition of Toxic Drugs and Chemicals in Man, 4th edn., pp. 12-15. Foster City, CA, USA: Chemical Toxicology Institute.
9. Bourne David W.A., Multiple IV Bolus Dose, boomer.org, Nov 28 2015

APPENDICES

Python program for Figure 5.2.2 page

```
# Program      : Euler's method for the system

import numpy as np
import matplotlib.pyplot as plt

print("Solution for dMb/dt = d1*[(Mg/Vg)-(Mb/Vb)]-d2*(Mb/Vb), dMg/dt = d1[(Mg/Vg)-
(Mb/Vb)]")

# Initializations

Dt = 0.015625                  # timestep Delta t
Mb_init = 0                   # initial Quantity of medicine in Blood
Mg_init = 325                  # initial Quantity of medicine in GI tract
t_init = 0                   # initial time
t_end = 60                    # stopping time

n_steps = int(round((t_end-t_init)/Dt)) # total number of timesteps

X = np.zeros(2)                # create space for current X=[Mg,Mb]^T
dXdt = np.zeros(2)              # create space for current derivative
t_arr = np.zeros(n_steps + 1)   # create a storage array for t
X_arr = np.zeros((2,n_steps+1))  # create a storage array for X=[Mg,Mb]^T
t_arr[0] = t_init             # add the initial t to the storage array
X_arr[0,0] = Mg_init            # add the initial Mg to the storage array
X_arr[1,0] = Mb_init            # add the initial Mb to the storage array

# Euler's method

for i in range (1, n_steps + 1):
   t = t_arr[i-1]             # load the time
   Mg = X_arr[0,i-1]            # load the value of Mg
   Mb = X_arr[1,i-1]            # load the value of Mb
   X[0] = Mg                # fill current state vector X=[Mg,Mb]^T
   X[1] = Mb
   dMbdt = 0.9*Mg-1.18*Mb        # calculate the derivative dMb/dt
   dMgdt = 0.98*Mb-0.9*Mg        # calculate the derivative dMg/dt
   dXdt[0] = dMgdt             # fill derivative vector dX/dt
   dXdt[1] = dMbdt
   Xnew = X + Dt*dXdt            # calculate X on next time step
   X_arr[:,i] = Xnew            # store Xnew
   t_arr[i] = t + Dt           # store new t-value


Dt = 0.015625
Mb_init = 125
Mg_init = 200
t_init = 0
```

```
t_end = 60

X = np.zeros(2)
dXdt = np.zeros(2)
t_ssx = np.zeros(n_steps + 1)
X_ssx = np.zeros((2,n_steps+1))
t_ssx[0] = t_init
X_ssx[0,0] = Mg_init
X_ssx[1,0] = Mb_init

for i in range (1, n_steps + 1):
    t = t_ssx[i-1]
    Mg = X_ssx[0,i-1]
    Mb = X_ssx[1,i-1]
    X[0] = Mg
    X[1] = Mb
    dMbdt = 0.9*Mg-1.18*Mb
    dMgdt = 0.98*Mb-0.9*Mg
    dXdt[0] = dMgdt
    dXdt[1] = dMbdt
    Xnew = X + Dt*dXdt             # calculate X on next time step
    X_ssx[:,i] = Xnew             # store Xnew
    t_ssx[i] = t + Dt             # store new t-value

Dt = 0.015625
Mb_init = 290                   # initial Quantity of Mb
Mg_init = 35                    # initial Quantity of Mg
t_init = 0                      # initial time
t_end = 60                      # stopping time

X = np.zeros(2)                  # create space for current X=[Mbg,Mb]^T
dXdt = np.zeros(2)                # create space for current derivative
t_llk = np.zeros(n_steps + 1)       # create a storage array for t
X_llk = np.zeros((2,n_steps+1))     # create a storage array for X=[Mg,Mb]^T
t_llk[0] = t_init                 # add the initial t to the storage array
X_llk[0,0] = Mg_init              # add the initial Mg to the storage array
X_llk[1,0] = Mb_init              # add the initial Mb to the storage array

for i in range (1, n_steps + 1):
    t = t_llk[i-1]                # load the time
    Mg = X_llk[0,i-1]             # load the value of Mb
    Mb = X_llk[1,i-1]             # load the value of Mg
    X[0] = Mg                     # fill current state vector X=[Mg,Mb]^T
    X[1] = Mb
    dMbdt = 0.9*Mg-1.18*Mb         # calculate the derivative dMb/dt
    dMgdt = 0.98*Mb-0.9*Mg         # calculate the derivative dMg/dt
    dXdt[0] = dMgdt               # fill derivative vector dX/dt
    dXdt[1] = dMbdt
    Xnew = X + Dt*dXdt             # calculate X on next time step
    X_llk[:,i] = Xnew             # store Xnew
    t_llk[i] = t + Dt             # store new t-value
```

```
# Plot the results

fig = plt.figure()
plt.plot(X_arr[0], X_arr[1], linewidth = 4, label="Mb=0,Mg=325")
plt.plot(X_ssx[0], X_ssx[1], linewidth = 4, label="Mb=125,Mg=200")
plt.plot(X_llk[0], X_llk[1], linewidth = 4, label="Mb=290,Mg=35")


plt.xlabel('Mg(t)', fontsize = 20)      # name of horizontal axis
plt.ylabel('Mb(t)', fontsize = 20)      # name of vertical axis

plt.xticks(fontsize = 15)               # adjust the fontsize
plt.yticks(fontsize = 15)               # adjust the fontsize
plt.axis([0, 500, 0, 350])              # set the range of the axes

plt.legend(fontsize=15)                 # show the legend
plt.show()                              # necessary for some platforms

# save the figure as .jpg (other formats: png, pdf, svg, (ps, eps))
fig.savefig('Combination.jpg', dpi=fig.dpi, bbox_inches = "tight")
```

Python program for Figure 5.1.1 page

```
# Program      : Euler's method for a system

import numpy as np
import matplotlib.pyplot as plt


Dt = 0.015625
Mb_init = 0
Mg_init = 325
t_init = 0
t_end = 900
Pill = 325

n_steps = int(round((t_end-t_init)/Dt))
X = np.zeros(2)
dXdt = np.zeros(2)
t_arr = np.zeros(n_steps + 1)
X_arr = np.zeros((2,n_steps+1))
t_arr[0] = t_init
X_arr[0,0] = Mg_init
X_arr[1,0] = Mb_init


for i in range (1, n_steps + 1):
    t = t_arr[i-1]              # load the time
    Mb = X_arr[1,i-1]
    Mg = X_arr[0,i-1]
```

```
    X[0] = Mg
    X[1] = Mb
    dMbdt = 0.9*Mg-1.23*Mb
    dMgdt = 0.98*Mb-0.9*Mg
    dXdt[0] = dMgdt                # fill derivative vector dX/dt
    dXdt[1] = dMbdt
    Xnew = X + Dt*dXdt             # calculate X on next time step
    X_arr[:,i] = Xnew            # store Xnew
    t_arr[i] = t + Dt            # store new t-value

if np.remainder(t,240)<0.005:
    Mg = Mg +325


# Plot the results

fig = plt.figure()
plt.plot(t_arr, X_arr[0,:], linewidth = 4, label="Mg(t)")
plt.plot(t_arr, X_arr[1,:], linewidth = 2, label="Mb(t)")  #

plt.xlabel('t (in minutes)', fontsize = 20)   # name of horizontal axis
plt.ylabel('Mb(t) and Mg(t)', fontsize = 20) # name of vertical axis

plt.xticks(fontsize = 15)              # adjust the fontsize
plt.yticks(fontsize = 15)              # adjust the fontsize
plt.axis([0, 1000, 0, 400])                # set the range of the axes

plt.legend(fontsize=15)                # show the legend
plt.show()                          # necessary for some platforms

# save the figure as .jpg (other formats: png, pdf, svg, (ps, eps))
fig.savefig('System.jpg', dpi=fig.dpi, bbox_inches = "tight")
```